



FLEET & SHORE PERSONNEL TRAINING

Effective Anchoring Training

Ground tackle, anchorage selection, a disciplined anchoring procedure, and how the bridge team keeps a safe anchor watch in all conditions.

SOLAS Ch. V

COLREG Rule 30

Company SMS

What We Will Cover

01

Anchoring Fundamentals

Ground tackle components and how an anchor holds

02

Selecting an Anchorage

Holding ground, depth, swinging room and shelter

03

The Anchoring Procedure

Preparation, approach and letting go, step by step

04

Anchor Watch & Monitoring

Keeping a disciplined watch and detecting dragging early

05

Weighing Anchor

Heaving in safely and reporting the anchor clear

06

Emergency Situations

Fouled anchor, dragging and heavy-weather response



SECTION 1

Anchoring Fundamentals

The equipment that holds a ship in place, and the basic physics of how it does so.

01

A Core Seamanship Skill

Anchoring safely and effectively is one of the most fundamental — and most frequently mishandled — evolutions in ship handling.

- A ship anchors to wait for a berth, a pilot, tide or cargo readiness, to shelter from weather, or as an emergency measure to avoid grounding or collision.
- An anchor does not hold a ship by weight alone — it holds primarily because the flukes dig into the seabed and the weight of chain lying on the bottom absorbs the ship's pull.
- A poorly planned or poorly executed anchoring evolution is one of the more common causes of groundings, collisions and damage to ground tackle in port waters.
- Every officer of the watch and every deck rating involved in anchor operations should understand both the equipment and the seamanship behind it, not just the commands.
- This training refreshes that understanding for all vessel types and applies equally to a single anchor or both-anchor moor.



Plan Before You Let Go

Every anchoring evolution should be planned and briefed in advance, the same way a berthing or pilotage passage is — not improvised on arrival.

GROUND TACKLE

Components of the Ground Tackle

"Ground tackle" refers to the complete system of equipment used to anchor the ship.



Anchor

Stockless bower anchors are standard on most merchant ships; flukes bite into the seabed to generate holding power.



Anchor Chain (Cable)

Made up in shackles (typically 27.5m/15 fathoms each), marked at each shackle joint so the amount paid out can be read at a glance.



Windlass

The deck machinery that heaves in and pays out the chain, and that holds the chain via the brake and bow-stopper when at anchor.



Hawse Pipe & Chain Locker

The hawse pipe guides the chain through the hull; the chain locker below stows it and lets it flake out freely when paying out.

Marking the Anchor Cable

Standard cable markings let the bridge and forecastle team confirm the amount of chain paid out without guesswork.

Shackles Out	Marking	Where Marked
1 shackle (27.5m)	One turn of wire / one link painted white	First link of the next shackle
2 shackles (55m)	Two turns of wire / two links painted white	Second link either side of the marker
3 shackles (82.5m)	Three turns of wire / three links painted white	Third link either side of the marker
Each successive shackle	One additional turn/link per shackle number	Studs on either side painted for visibility



SECTION 2

Selecting an Anchorage

Choosing where to anchor is as important to a safe evolution as how the anchor is let go.

02

What Makes an Anchorage Suitable

The chart, the pilot book and local knowledge together inform a sound anchorage choice.

- Holding ground should be good — sand, mud or clay generally hold well; rock, hard shingle or heavy weed give poor and unpredictable holding.
- Depth should suit the amount of chain the ship can reasonably deploy, allowing an adequate scope without dragging the chain locker empty.
- Swinging room must be sufficient for the ship to swing through a full circle around the anchor as the tide and wind change, clear of other vessels, shoals and obstructions.
- Shelter from the prevailing and forecast wind and sea reduces both the load on the ground tackle and the risk of dragging.
- Designated or customary anchorage areas shown on the chart should be used wherever available, respecting any local port or VTS anchoring instructions.



Under-Keel Clearance

Confirm adequate under-keel clearance at all states of tide throughout the intended stay, not just at the moment of anchoring.

SCOPE OF CABLE

How Much Chain to Use

Scope — the ratio of chain length paid out to depth of water — is the single biggest factor in secure holding.

4-6x

Typical scope (chain length to depth) for moderate weather and good holding ground

6-8x

Increased scope recommended for strong winds, poor holding ground or an exposed anchorage

8-10x+

Heavy-weather scope, used together with a second anchor where conditions and swinging room allow

Assessing the Anchorage Before Arrival

A few minutes of chart planning avoids most anchoring problems before the anchor ever touches the seabed.



Do

- Check the chart for holding ground, foul ground, cables and pipelines in the area
- Calculate swinging circle against charted depths and the position of other anchored ships
- Confirm the anchorage with VTS or port control where required
- Brief the bridge team and forecastle party on the planned position, depth and scope



Don't

- Don't anchor on a charted cable, pipeline or wreck without express local authorisation
- Don't anchor closer to other vessels than swinging room safely allows
- Don't rely on electronic charts alone without cross-checking soundings and local notices
- Don't commit to an anchorage without a clear plan for what happens if the wind or swell builds



SECTION 3

The Anchoring Procedure

A disciplined, step-by-step evolution from preparation to letting go.

03

BEFORE ARRIVAL

Preparing to Anchor

Preparation begins well before the ship reaches the anchorage.

- The Master briefs the bridge team on the anchoring plan: position, expected depth, scope to be used and any contingencies.
- The forecastle party is briefed and mustered in good time, in full PPE, with clear communication established with the bridge (radio and hand signals).
- The windlass is tested, the brake checked, and the anchor is walked back clear of the hawse pipe ready for letting go.
- Speed is reduced progressively on the approach so the ship carries minimal way over the ground at the moment of letting go.
- The Master confirms position against the planned anchorage using all available means — GPS, radar ranges and bearings, and visual transits.



Clear Communication

The forecastle party and bridge must maintain continuous, unambiguous communication throughout — confirming each order given and each action completed.

STEP BY STEP

Letting Go the Anchor

A standard sequence keeps the evolution controlled and repeatable.

Step	Action	Key Check
1	Ship stopped or making minimal headway over the ground at the chosen position	Confirm position and depth
2	Anchor let go and allowed to run to the seabed under control of the brake	Chain paying out smoothly, not surging
3	Ship allowed to gather sternway or drift back as chain is paid out to the planned scope	Monitor chain angle and load
4	Chain brought to short stay, brake applied and windlass secured; anchor reported holding	Confirm position steady on fixes

Anchor Party Discipline

The forecastle is a high-energy work area during anchoring — discipline there prevents injury.



Do

- Wear full PPE including hard hat, safety boots and gloves at all times
- Stand clear of the chain's bight and the direct line of the cable when it is running
- Keep hands and feet clear of the windlass gypsy and chain stopper
- Report chain markings and rate of run to the bridge clearly and promptly



Don't

- Don't stand in the direct line of the chain when it is running out under load
- Don't attempt to clear a jam on a moving windlass without stopping it first
- Don't rely on the brake alone without engaging the bow-stopper once anchored
- Don't leave the forecastle unattended until the anchor is confirmed holding



SECTION 4

Anchor Watch & Monitoring

04

The anchor holding is confirmed once — keeping it holding is a continuous responsibility.

What an Effective Anchor Watch Does

An anchor watch is a full bridge watch, not a reduced one — the ship remains the watchkeeper's full responsibility.

- Position is checked and plotted at regular, defined intervals — more frequently in poor holding ground, strong tide or deteriorating weather.
- Anchor bearings, GPS position and radar range/bearing to a fixed reference point are all used together to detect the smallest sign of movement.
- Weather and sea state are monitored continuously, with the Master informed of any deterioration well before it becomes critical.
- Main engine readiness is maintained per the Company's SMS and port requirements, so propulsion is available promptly if needed.
- A proper lookout is maintained at all times, including for other vessels manoeuvring, drifting or anchoring nearby.



Set an Anchor Alarm

Modern ECDIS and GPS anchor-watch alarms should be set to the calculated swinging radius as a backup to visual and manual position checks, never as a substitute for them.

RECOGNISING THE SIGNS

Detecting a Dragging Anchor

Early detection turns a dragging anchor from an emergency into a routine correction.

Indicator	What to Check	Typical Cause
Position drift on fixes	Successive GPS/radar fixes show steady movement off the anchor position	Insufficient scope or poor holding ground
Changing chain angle/tension	Chain leads unusually taut, or angle to the bow changes without tide change	Anchor breaking out under load
Erratic swinging	Ship fails to swing normally with tide/wind, or swings beyond the calculated radius	Anchor dragging or fouled
Vibration felt through the chain	A rhythmic 'snubbing' or jerking felt at the windlass or hull	Anchor dragging over a hard or uneven bottom



SECTION 5

Weighing Anchor

Getting under way again is as much a controlled evolution as anchoring itself.

05

Heaving In Safely

A controlled, well-communicated heave-in avoids damage to the ground tackle and keeps the bridge fully informed throughout.

- The Master briefs the bridge team and forecastle party in advance, including the planned departure track and any traffic in the vicinity.
- The windlass heaves in steadily; the forecastle party reports chain markings, direction of pull and the angle of the chain ("up and down", "short stay") as it comes in.
- Engine and helm are used as needed to ease the load on the windlass and bring the ship over the anchor, rather than heaving against a taut chain.
- Once the anchor is aweigh (clear of the bottom), the ship is officially under way and subject to the navigation rules for a vessel making way.
- The anchor and chain are washed down as they come aboard to prevent mud and debris being carried into the chain locker.



Report 'Anchor Aweigh'

The forecastle party's report that the anchor is aweigh and clear of the water is the formal signal the ship is free to manoeuvre — log the time precisely.

FINAL CHECKS

Before Reporting 'Anchor Home'

A short final check confirms the ground tackle is properly secured for sea.



Anchor Fully Housed

Anchor seated correctly in the hawse pipe, with flukes aligned and no fouling on the hull.



Chain Stopper & Lashing Secured

Bow-stopper engaged and any sea-lashing rigged for the intended passage and expected weather.



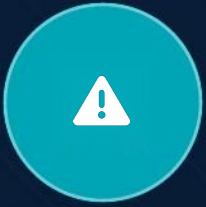
Damage Check

Anchor, hawse pipe and windlass visually checked for damage sustained during the anchorage.



Report to Bridge

Forecastle party formally reports "anchor home and secured for sea" before the watch stands down.



SECTION 6

Emergency Situations

When anchoring does not go to plan — recognising and responding to the two most common anchor emergencies.

06

Responding to a Dragging Anchor

Once dragging is confirmed, prompt, decisive action prevents a routine problem becoming a grounding or collision.



Do

- Inform the Master immediately and sound the appropriate internal alarm per the SMS
- Pay out additional cable if depth and searoom allow, to increase holding power
- Start the main engine and stand by to manoeuvre or re-anchor
- Monitor position continuously and warn nearby vessels by VHF if a close-quarters situation is developing



Don't

- Don't wait to 'see if it settles' once dragging is confirmed on fixes
- Don't attempt to weigh anchor in heavy weather without adequate engine power ready
- Don't ignore an anchor alarm without personally verifying the ship's position
- Don't delay reporting a dragging anchor to the Company where required by the SMS

Dealing With a Fouled Anchor

An anchor can foul on its own chain, on a wreck, cable or an obstruction on the seabed, or on another ship's cable in a crowded anchorage.

- Signs of a fouled anchor include unusual heaving-in resistance, a chain leading in an unexpected direction, or visible fouling as the anchor breaks the surface.
- Never apply excessive force with the windlass against a suspected fouled anchor — this risks damaging the windlass, the chain or the hull.
- Manoeuvring the ship slowly around the fouled point, under engine and helm, can sometimes clear a foul without further intervention.
- Where the anchor is fouled on another vessel's chain, communication with that vessel and, if necessary, the port authority is essential before attempting to clear it.
- A fouled anchor that cannot be cleared safely should be reported to the Master, the Company and, where relevant, the port authority without delay.



Stop and Assess

The first response to any unexpected resistance while weighing anchor is to stop heaving and assess — forcing it risks far more damage than the delay of a careful diagnosis.

KEY TAKEAWAYS

What Every Bridge Team Should Remember

Three principles that apply across every topic in this course.

Plan

Treat every anchoring evolution with the same planning and briefing as a berthing or pilotage passage

Watch

Maintain a full, disciplined anchor watch — position checks, weather monitoring and engine readiness

Act Early

Recognise dragging or fouling early and respond decisively, before it becomes an emergency



COURSE COMPLETE

A well-planned, well-watched anchorage is one of the safest places a ship can be — a poorly managed one is one of the riskiest.

Company DPA — 24/7 emergency line

Nearest RCC / MRCC — via VHF Ch.16 / DSC

Fleet Safety Department — mng@gloryship.com.tr